Independent University, Bangladesh

CSE313 Compiler Construction · Practice Set C · Mid + Quiz · Time: 1h30m

New questions. Q1 is NOT ambiguous (like Aut-23). Q2 is NOT LL(1). Q3 is a reason question (like Summer-23).

Question paper

Mid Question (Total marks: 20)

1. Prove whether the following grammar is Ambiguous or NOT. If the grammar is ambiguous, convert it to unambiguous form. **[7]**

```
E → E $ T | T
T → F % T | F
F → F ! | ( E ) | n
```

2. Write the First() and Follow() of the following Grammar. Draw LL(1) Parsing Table and parse the string `a a b` showing all steps. If parsing is not possible, point out the reasons. **[8]**

```
S → A a B | b C
A → a A | ε
B → b | ε
C → c C | d
```

3. A student removed left recursion from an ambiguous grammar and claimed the new grammar is unambiguous. Is the claim correct? Justify with an example. **[5]**

Quiz Question (Total marks: 10)

1. Convert the following Non-Deterministic Grammar (NDG) to Deterministic Grammar (DG): **[5]**

```
X → p q r X | p q r Y | p q | p s
Y → y
```

2. Convert the following Left Recursive Grammar (LRG) to Right Recursive Grammar (RRG): **[5]**

```
P → P + Q | P - Q | Q
Q → Q * id | id
```